Chapter 2

Basic notions of representation theory

2.1. What is representation theory?

In technical terms, representation theory studies representations of associative algebras. Its general content can be very briefly summarized as follows.

An **associative algebra** over a field k is a vector space A over k equipped with an associative bilinear multiplication $a, b \mapsto ab$, $a, b \in A$. We will always consider associative algebras with unit, i.e., with an element 1 such that $1 \cdot a = a \cdot 1 = a$ for all $a \in A$. A basic example of an associative algebra is the algebra $\mathrm{End}V$ of linear operators from a vector space V to itself. Other important examples include algebras defined by generators and relations, such as group algebras and universal enveloping algebras of Lie algebras.

A **representation** of an associative algebra A (also called a **left A-module**) is a vector space V equipped with a homomorphism $\rho : A \to \mathrm{End}V$, i.e., a linear map preserving the multiplication and unit.

A **subrepresentation** of a representation V is a subspace $U \subset V$ which is invariant under all operators $\rho(a)$, $a \in A$. Also, if V_1, V_2 are two representations of A, then the **direct sum** $V_1 \oplus V_2$ has an obvious structure of a representation of A.